

miR-181a/b inhibition protects against glaucomatous optic neuropathy by enhancing mitochondrial resilience and reducing neuroinflammation

Background/Introduction

Glaucoma is a leading cause of irreversible blindness characterized by retinal ganglion cell (RGC) loss and optic nerve degeneration. Current therapies target intraocular pressure (IOP), yet neurodegeneration often progresses through IOP-independent mechanisms. Mitochondrial dysfunction and oxidative stress contribute to RGC vulnerability. We previously identified microRNAs 181a and b (miR-181a/b) as regulators of mitochondrial biogenesis and function, and showed that their downregulation is neuroprotective in mitochondrial optic neuropathy models. Here, we investigated whether miR-181a/b inhibition protects against glaucomatous optic neuropathy.

Methods

Two mouse models were employed: optic nerve crush (ONC) to model acute axonal injury and DBA/2J mice as a chronic high-tension glaucoma model. ONC was performed in miR-181a/b^{+/+} and miR-181a/b^{-/-} mice at 4 months of age. In DBA/2J mice, a single intravitreal injection of adeno-associated viral (AAV) vectors expressing a miR-181a/b inhibitory sponge or GFP control was administered at defined stages. Functional assessment was performed by photopic negative response (PhNR). Structural and molecular analyses included RGC quantification, optic nerve histology, axonal density, mitochondrial content, and inflammatory marker profiling.

Results

In both models, miR-181a/b inactivation resulted in significant neuroprotection. In ONC, genetic deletion preserved RGC survival and visual function (OKR) compared to controls. In DBA/2J mice, AAV-mediated inhibition improved PhNR responses, increased RGC survival, preserved optic nerve architecture, and enhanced axonal density. GFP expression extended along the optic nerve up to 7 months after a single injection, indicating durable vector persistence. Treated optic nerves displayed increased mitochondrial abundance and reduced inflammatory marker intensity, consistent with improved mitochondrial integrity and reduced neuroinflammation.

Conclusions

miR-181a/b inhibition provides sustained neuroprotection in experimental glaucoma by enhancing mitochondrial resilience and reducing inflammation. These findings identify miR-181a/b as a promising RNA-based therapeutic target for glaucomatous optic neuropathy and related neurodegenerative disorders characterized by mitochondrial vulnerability.